

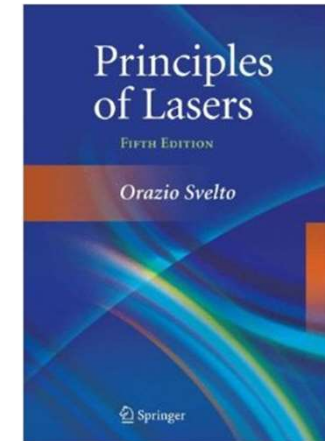


南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《激光原理与技术》

Principles and Technologies of Lasers

Ch2-6 Open Cavity & Gaussian Beam



Dr. Dan Luo(罗丹)

Email: luod@sustech.edu.cn

Office: Room 331 at College
of Engineering South Tower

Tel: 0755-88018552



Chapter 2: Open Cavity & Gaussian Beam

❖ 2.11 The Focus and Alignment of Gaussian beam

- The focus of Gaussian beam
- The alignment of Gaussian beam

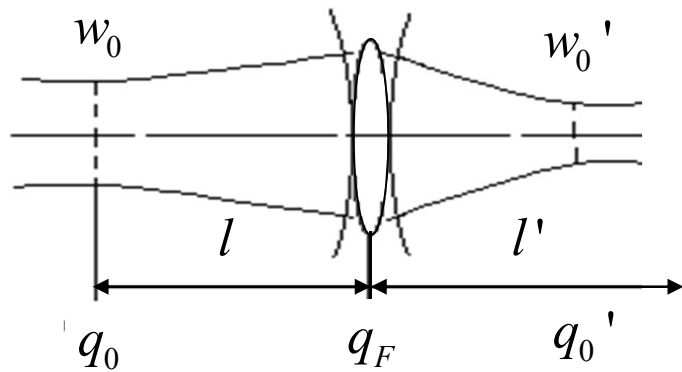
❖ 2.13 Beam diffraction magnification factor

❖ 2.14 Unstable resonator (not required)

- Formation of the unstable resonator
- Conjugate image point and self-reconstructed mode
- Geometrical magnification
- Energy loss

2.11 The focus & alignment of Gaussian beam

- **Focus:** Increase the power density of laser beam
- **Alignment:** Decrease the laser divergence angle, increase the energy transfer distance

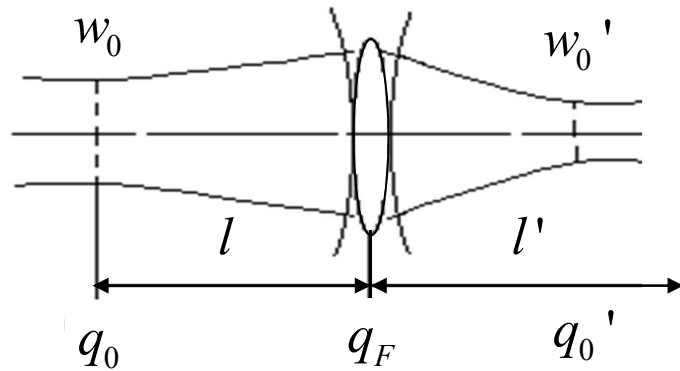


$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\frac{1}{F} & 1 \end{pmatrix} \begin{pmatrix} 1 & l \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & l \\ -\frac{1}{F} & 1 - \frac{l}{F} \end{pmatrix}$$

$$q_F = \frac{Aq_0 + B}{Cq_0 + D} = \frac{i\frac{\pi w_0^2}{\lambda} + l}{-\frac{1}{F}i\frac{\pi w_0^2}{\lambda} + \left(1 - \frac{l}{F}\right)} = i\frac{\pi w_0'^2}{\lambda} - l'$$

$$l' = F + \frac{(l-F)F^2}{(l-F)^2 + \left(\frac{\pi w_0^2}{\lambda}\right)^2} \quad w_0'^2 = \frac{F^2 w_0^2}{(l-F)^2 + \left(\frac{\pi w_0^2}{\lambda}\right)^2}$$

The focus of Gaussian beam $w_0' < w_0$



$$w_0'^2 = \frac{F^2 w_0^2}{(l - F)^2 + \left(\frac{\pi w_0^2}{\lambda} \right)^2}$$

1. F is fixed, 1) When $l < F$, if l decrease, beam size decrease, minimum at $l=0$

In order to have focus, we need to have $F^2 < (l - F)^2 + f^2$

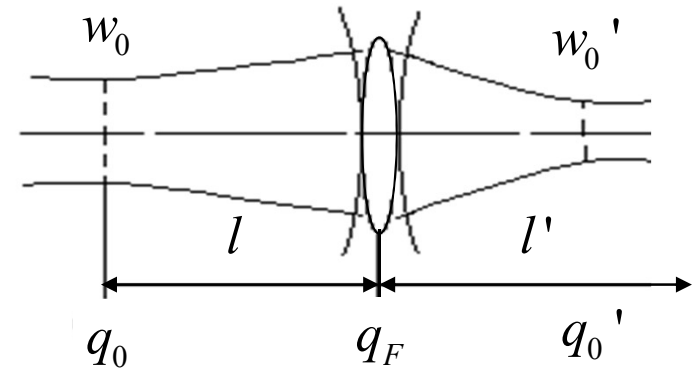
That is $l > F + \sqrt{F^2 - f^2}$ or $l < F - \sqrt{F^2 - f^2}$

There will be no focus effect in the range of

$$F - \sqrt{F^2 - f^2} < l < F + \sqrt{F^2 - f^2}$$

The focus of Gaussian beam $w_0' < w_0$

$$w_0'^2 = \frac{F^2 w_0^2}{(l-F)^2 + \left(\frac{\pi w_0^2}{\lambda}\right)^2}$$



1. F is fixed, 1) When $l < F$, if l decrease, beam size decrease, minimum at $l=0$

$$w_0' = \frac{w_0}{\sqrt{\left(\frac{l-F}{F}\right)^2 + \left(\frac{\pi w_0^2}{\lambda F}\right)^2}} \quad \text{if } l=0, \quad (w_0')_{\min} = \frac{w_0}{\sqrt{1 + \left(\frac{f}{F}\right)^2}}$$

Then we have

$$l' = F + \frac{(l-F)F^2}{(l-F)^2 + \left(\frac{\pi w_0^2}{\lambda}\right)^2} = F \left[1 - \frac{1}{1 + \left(\frac{f}{F}\right)^2} \right] < F$$

$$K = \frac{w_0'}{w_0} = \frac{1}{\sqrt{1 + \left(\frac{f}{F}\right)^2}} < 1$$

Has focus effect for any F , and the position is within F

1. F is fixed, 2) When $l > F$, beam size decrease, for $l \rightarrow \infty$

$$w_0'^2 = \frac{F^2 w_0^2}{(l-F)^2 + \left(\frac{\pi w_0^2}{\lambda}\right)^2} \rightarrow 0 \quad l' = F + \frac{(l-F)F^2}{(l-F)^2 + \left(\frac{\pi w_0^2}{\lambda}\right)^2} \rightarrow F$$

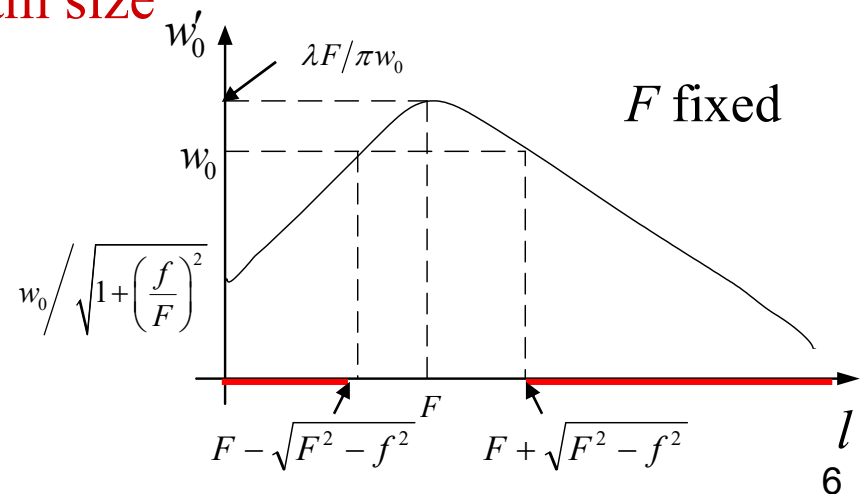
$$\text{for } l \gg F \quad \frac{1}{w_0'^2} = \frac{(l-F)^2 + \left(\frac{\pi w_0^2}{\lambda}\right)^2}{w_0^2 F^2} = \frac{1}{w_0^2} \left(\frac{l}{F}\right)^2 + \frac{1}{F^2} \left(\frac{\pi w_0}{\lambda}\right)^2 = \frac{\pi^2}{F^2 \lambda^2} w^2(l)$$

$$\text{then } w_0' \approx \frac{\lambda}{\pi w(l)} F, \text{ and } l' = F$$

For bigger l and smaller F , we will have smaller beam size

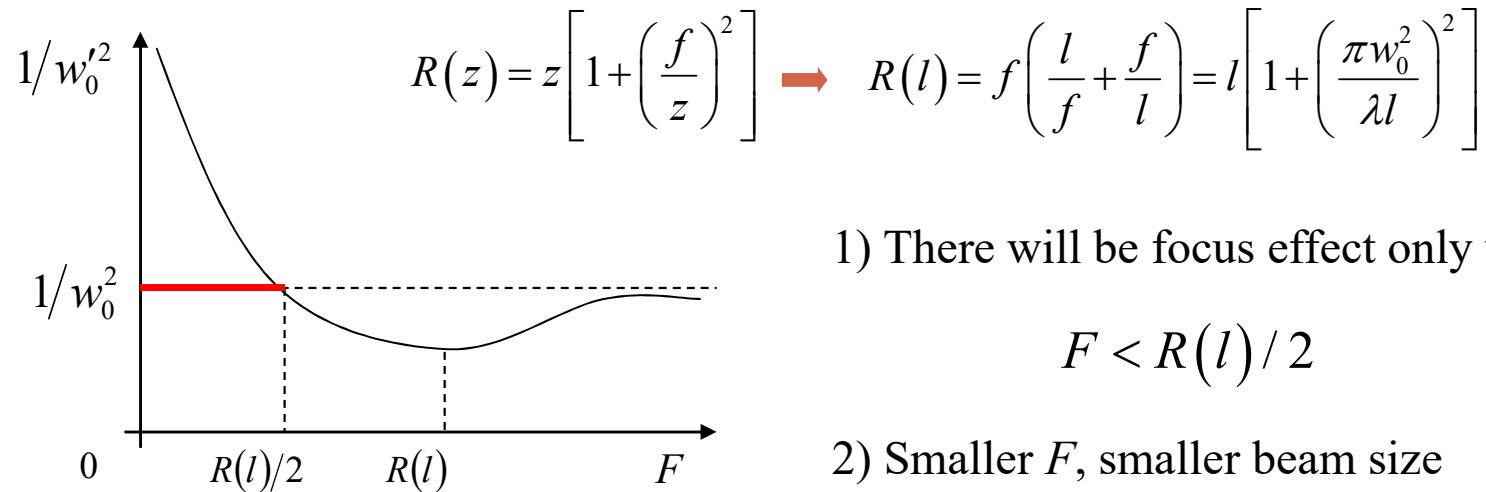
1. F is fixed, 3) When $l=F$, maximum beam size

$$w_0'^2 = \frac{F^2 w_0^2}{(l-F)^2 + \left(\frac{\pi w_0^2}{\lambda}\right)^2} = \left(\frac{\lambda}{\pi w_0} F\right)^2$$



2. l is fixed

$$\frac{1}{w_0'^2} = \frac{1}{w_0^2} \left[\left(\frac{l-F}{F} \right)^2 + \left(\frac{f}{F} \right)^2 \right]$$



1) There will be focus effect only when

$$F < R(l) / 2$$

2) Smaller F , smaller beam size

Example: Wavelength of the Gaussian beam is $3.14 \mu\text{m}$ and the minimum beam size is 1 mm . If put the lens on the beam waist and 2 m away from the beam waist, respectively, please determine the focal length of the lens to have focus effect.

$$f = \frac{\pi w_0^2}{\lambda} = \frac{3.14 \times 10^{-6}}{3.14 \times 10^{-6}} = 1$$

1) For $l=0$, any lens is fine

2) For $l=2$, $\frac{w_0'}{w_0} = \frac{F}{\sqrt{(l-F)^2 + f^2}} < 1$, we have $F < 1.25 \text{ m}$

The alignment of Gaussian beam

1. The situation for single lens

$$\theta_0 = 2 \frac{\lambda}{\pi w_0} \quad \theta'_0 = 2 \frac{\lambda}{\pi w'_0}$$

$$w'_0 > w_0 \quad \theta'_0 < \theta_0$$

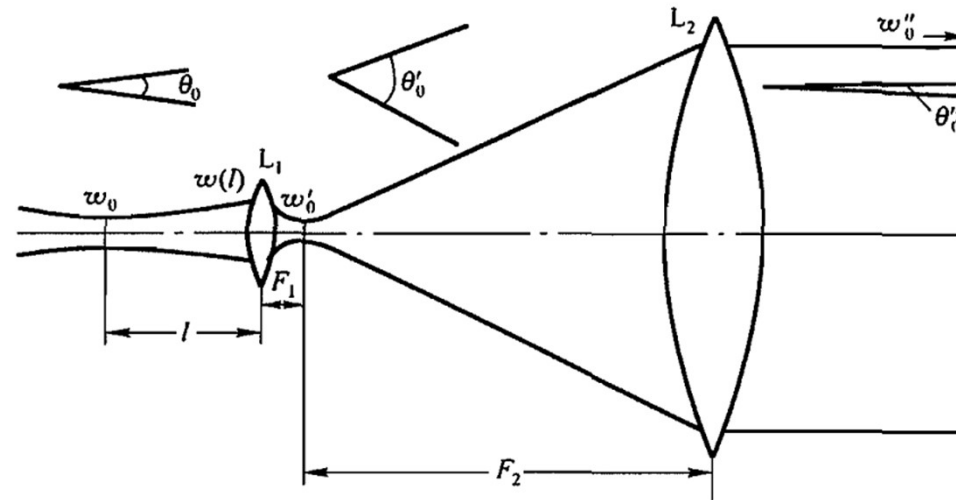
From $\partial \frac{1}{w_0'^2} / \partial l = 0$ We know when $l = F$, $(w_0)_{\max} = \frac{\lambda}{\pi w_0} F$

Bigger F and **smaller w_0** will have better performance

$$w_0'^2 = \frac{w_0^2 F^2}{(F-l)^2 + \left(\frac{\pi w_0}{\lambda}\right)^2} \Rightarrow \theta'_0 = \frac{2\lambda}{\pi} \sqrt{\frac{1}{w_0^2} \left(1 - \frac{l}{F}\right)^2 + \frac{1}{F^2} \left(\frac{\pi w_0}{\lambda}\right)^2}$$

For specific w_0 , whatever F and l , it can not have infinite beam size. Therefore, it is **impossible** to convert Gaussian beam into plane wave (i.e. good alignment) by using a **single lens**.

2. The application of inverted telescope



Collimation ratio $M' = \frac{\theta_0}{\theta_0''}$

$$\theta_0 = 2 \frac{\lambda}{\pi w_0} \quad \theta_0' = 2 \frac{\lambda}{\pi w_0'} = 2 \frac{w_0}{F} \quad \frac{\theta_0''}{\theta_0'} = \frac{\pi w_0'^2}{\lambda F_2} \quad \frac{\theta_0'}{\theta_0} = \frac{2\lambda}{\pi w_0'} \bigg/ \frac{2\lambda}{\pi w_0} = \frac{w_0}{w_0'}$$

$$w_0' = \frac{\lambda F_1}{\pi w(l)} \quad \frac{\theta_0''}{\theta_0} = \left(\frac{\theta_0''}{\theta_0'} \right) \left(\frac{\theta_0'}{\theta_0} \right) = \frac{\pi}{\lambda F_2} w_0 w_0' = \frac{F_1 w_0}{F_2 w(l)}$$

$$M' = \frac{\theta_0}{\theta_0''} = \frac{F_2 w(l)}{F_1 w_0} = M \frac{w(l)}{w_0} = M \sqrt{1 + \left(\frac{l}{f} \right)^2} = M \sqrt{1 + \left(\frac{\lambda l}{\pi w_0^2} \right)^2}$$

2.13 Beam diffraction magnification factor

It is good to describe the laser beam quality by beam size and divergence angle, but those parameters will change after passing the optical component.

It is noted that $w_0\theta_0 = 2\lambda/\pi$

$$\left\{ \begin{array}{l} \text{For Hermit-Gaussian beam} \\ \text{For Laguerre-Gaussian beam} \end{array} \right. \quad \left\{ \begin{array}{l} w_m\theta_m = (2m+1)2\lambda/\pi \\ w_n\theta_n = (2n+1)2\lambda/\pi \\ w_{mn}\theta_{mn} = (m+2n+1)2\lambda/\pi \end{array} \right.$$

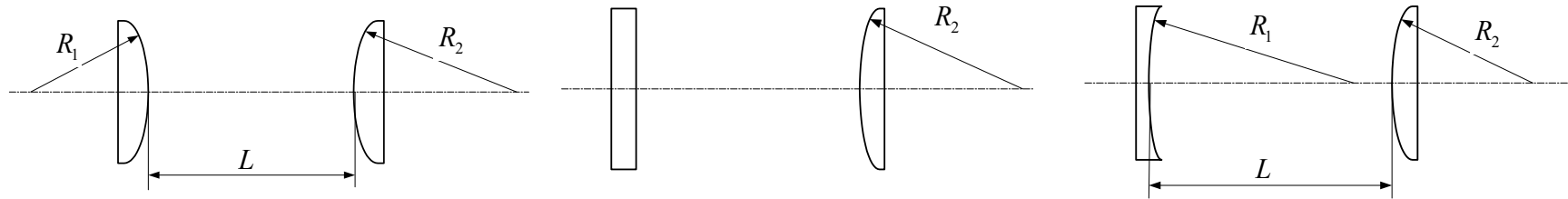


$$M^2 = \frac{\text{Beam Size} * \text{Divergence Angle}}{\text{Beam Size} * \text{Divergence Angle (of the TEM}_{00})}$$

$M^2 = 1$ Fundamental base mode	$M_x^2 = (2m+1)$ $M_y^2 = (2n+1)$ $M_r^2 = (m+2n+1)$	Different modes without coherence	$\left\{ \begin{array}{l} M_x^2 = \sum_m \sum_n (2m+1) C_{mn} ^2 \\ M_y^2 = \sum_m \sum_n (2n+1) C_{mn} ^2 \\ M_r^2 = \sum_m \sum_n (m+2n+1) C_{mn} ^2 \end{array} \right.$
	High-order mode		

2.14 Unstable Resonator

1. The formation of unstable resonator: high loss cavity



biconvex lens
双凸

Flat convex lens
平凸

Concave-convex lens
凹凸（虚共焦望远镜）

$$g_1 > 1, g_2 > 1, g_1 g_2 > 1$$

$$g_1 > 1, g_2 = 1, g_1 g_2 > 1$$

$$R_1 < L, R_1 + R_2 = R_1 - |R_2| > L$$

Characteristics:

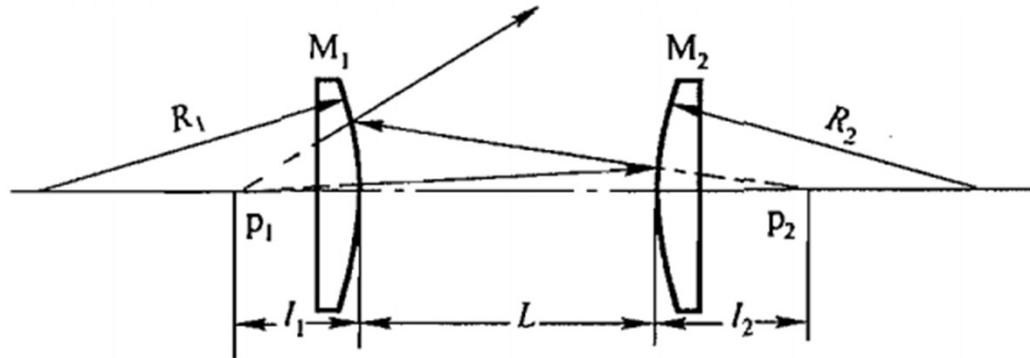
Transverse mode loss is high, the mode volume is big, small divergence angle,
Able to achieve high power high beam quality of single mode operation

Analysis approach:

geometrical optics; wave optics approximate analytical theory

One-trip loss $\gg$ diffraction loss, cavity only exists a pair of conjugated image point and the corresponding self-reconstructed mode (resonant mode)

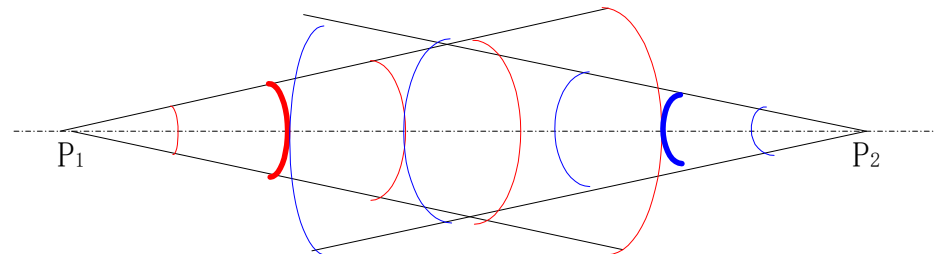
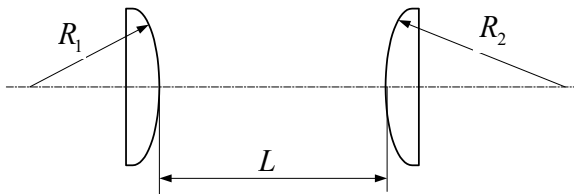
2. Conjugate image point and geometric self-reconstructed mode



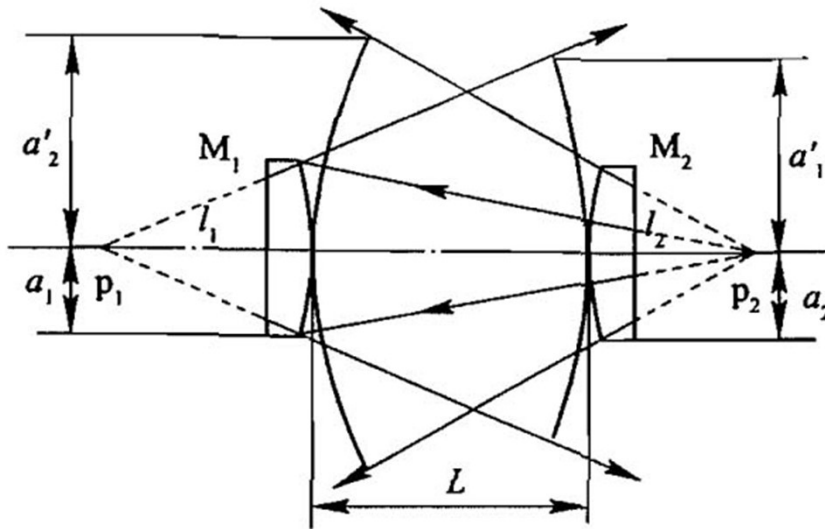
$$\begin{cases} \frac{1}{l_1 + L} - \frac{1}{l_2} = \frac{2}{R_2} \\ \frac{1}{l_2 + R} - \frac{1}{l_1} = \frac{2}{R_1} \end{cases}$$

$$l_1 = \frac{\pm \sqrt{L(L - R_1)(L - R_2)(L - R_1 - R_2)} - L(L - R_2)}{2L - R_1 - R_2}$$

$$l_2 = \frac{\pm \sqrt{L(L - R_1)(L - R_2)(L - R_1 - R_2)} - L(L - R_1)}{2L - R_1 - R_2}$$



3. Geometrical magnification



One-trip magnification

$$m_1 = a'_1/a_1 \quad m_2 = a'_2/a_2$$

Round trip magnification

$$M = m_1 m_2$$

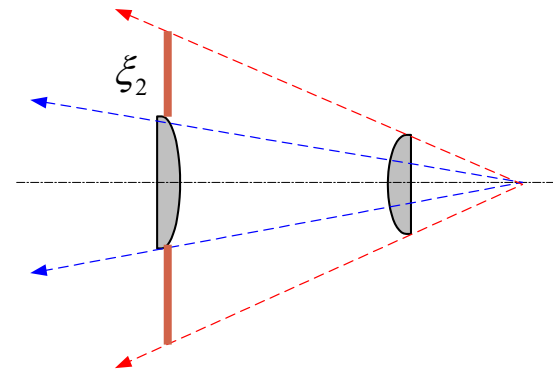
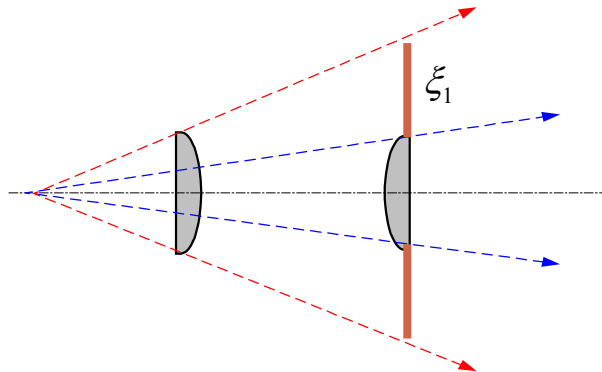
Double convex cavity

$$m_1 = \frac{l_1 + L}{l_1} \quad m_2 = \frac{l_2 + L}{l_2}$$

The telescope cavity (virtual confocal cavity, or real confocal cavity)

$$M = m_1 m_2 = \left| \frac{R_1}{R_2} \right|$$

4. Energy loss



$M_1 \rightarrow M_2$ The escape energy from M_2

$$\xi_1 = 1 - \Gamma_1 = 1 - \left(\frac{a_2}{a_1} \right)^2 / m_1^2$$

Energy reflected back by M_2

$M_2 \rightarrow M_1$ The escape energy from M_1

$$\xi_2 = 1 - \Gamma_2 = 1 - \left(\frac{a_1}{a_2} \right)^2 / m_2^2$$

Energy reflected back by M_1

where

$$\Gamma_1 = \left(\frac{a_2}{a'_1} \right)^2 = \left(\frac{a_2}{a_1 m_1} \right)^2 \quad \Gamma_2 = \left(\frac{a_1}{a'_2} \right)^2 = \left(\frac{a_1}{a_2 m_2} \right)^2$$

$$\xi_2 = 1 - \Gamma_1 \Gamma_2 = 1 - \frac{1}{m_1^2} \frac{1}{m_2^2} = 1 - \frac{1}{M^2}$$

$M = m_1 m_2$ — Magnification

The average one-trip loss

$$\xi_1 = 1 - \sqrt{\Gamma_1 \Gamma_2} = 1 - \frac{1}{M}$$

Example: $|R| = 10m, L = 1m$

Symmetrical biconvex $m = 1.558 \Rightarrow M = 2.428 \quad \xi_1 = 1 - (1/M) \approx 59\%$